

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 2025
Team ID	
Project Name	India's Agriculture Crop Production Analysis
Maximum Marks	4 Marks

Technical Architecture:

The architecture below shows the flow for the India's Agriculture Crop Production Analysis application: the user submits crop details through the Web UI, the application logic validates and forwards the request, historical crop data is fetched from the cloud database, the trained Machine Learning model predicts the crop production / yield, and the result is returned to the user and shown on the analytics dashboard.

USER → Web UI (Form: State / District / Season / Crop / Area) → Application Logic (Flask / Django) → Data Preprocessing → ML Prediction Model (Random Forest / Regression) → Cloud Database (historical crop dataset) → Predicted Production / Yield → Visualization (charts / dashboard) → ADMIN (dataset upload, model retraining, analytics).

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the application (Web UI)	HTML, CSS, JavaScript, Bootstrap
2.	Application Logic-1	Handles user input, request validation and routing	Python, Flask
3.	Application Logic-2	Data preprocessing and feature engineering on crop dataset	Python (Pandas, NumPy)
4.	Application Logic-3	Crop production / yield prediction using trained model	Scikit-learn (Random Forest / Linear Regression)
5.	Database	Storage of user data, crop records and prediction history	MySQL
6.	Cloud Database	Hosting of historical crop production dataset on cloud	IBM Db2 on Cloud / MySQL on Cloud
7.	File Storage	Storage of uploaded datasets and generated reports	Local Filesystem / IBM Cloud Object Storage
8.	External API-1	Fetching real-time rainfall / weather data for a region	IBM Weather API / OpenWeather API
9.	External API-2	Fetching soil / geographic data for a district (if applicable)	Government Open Data API (data.gov.in)
10.	Machine Learning Model	Predicts crop production based on historical trends	Random Forest Regressor / XGBoost
11.	Infrastructure (Server / Cloud)	Application Deployment Local Server Configuration: Flask Dev Server Cloud Server Configuration: IBM Cloud Foundry	Local, IBM Cloud Foundry

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used to build the application and model	Flask, Scikit-learn, Pandas, NumPy, Bootstrap
2.	Security Implementations	Access controls and data protection implemented	Password hashing (bcrypt/SHA-256), HTTPS, Session-based auth
3.	Scalable Architecture	Justification of scalability of the architecture	3-tier architecture (Presentation, Application, Data) supporting horizontal scaling
4.	Availability	Justification of application availability	Cloud hosting with load balancing to minimize downtime
5.	Performance	Design considerations for performance	Caching of frequent queries, optimized model inference, indexed database queries

References:

<https://c4model.com/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://data.gov.in/>